

NIZHNEVARTOVSK, Russian Federation

WMO: 234710

Lat: 60.949N

Lon: 76.484E

Elev: 54

StdP: 100.68

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.1	(c) -37.1	(d) -43.0	(e) 0.1	(f) -39.4	(g) -39.9	(h) 0.1	(i) -37.2	(j) 8.9	(k) -11.7	(l) 8.4	(m) -15.0	(n) 2.3	(o) 40	(p) 0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	28.6	19.2	26.9	18.7	24.9	17.9	20.9	26.1	19.8	24.5	18.7	23.3	4.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.0	23.1	18.0	13.0	22.2	16.9	12.2	21.2	60.8	26.3	56.9	24.8	53.3	23.5	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.9	(b) 8.2	(c) 7.4	(e) -41.1	(f) 31.3	(g) 4.0	(h) 1.6	(i) -44.0	(j) 32.4	(k) -46.4	(l) 33.4	(m) -48.7	(n) 34.3	(o) -51.6	(p) 35.5		
(a) 8.9	(b) 8.2	(c) 7.4	(e) -40.9	(f) 22.8	(g) 3.6	(h) 1.2	(i) -43.5	(j) 23.7	(k) -45.7	(l) 24.4	(m) -47.7	(n) 25.1	(o) -50.3	(p) 26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-21.4	-18.7	-7.9	-0.1	5.9	15.4	17.9	14.4	8.4	0.0	-11.3	-17.6	(6)
(7)		DBStd	14.98	9.60	8.48	6.42	6.00	5.48	5.21	3.87	3.75	4.34	5.61	8.63	9.30	(7)
(8)		HDD10.0	4696	972	804	556	305	155	13	1	4	81	311	640	854	(8)
(9)		HDD18.3	7194	1231	1037	814	553	388	116	57	131	297	568	890	1113	(9)
(10)		CDD10.0	627	0	0	0	2	27	176	246	142	34	2	0	0	(10)
(11)		CDD18.3	83	0	0	0	0	1	28	43	10	1	0	0	0	(11)
(12)		CDH23.3	721	0	0	0	0	23	271	360	65	2	0	0	0	(12)
(13)		CDH26.7	157	0	0	0	0	4	64	83	6	0	0	0	0	(13)
(14)	Wind	WSAvg	3.7	3.4	3.3	4.2	4.1	4.1	3.8	3.3	3.2	3.5	4.0	3.8	3.6	(14)
(15)	Precipitation	PrecAvg	793	48	41	54	51	65	76	71	85	68	87	82	66	(15)
(16)		PrecMax	1003	86	73	122	82	117	149	144	126	114	141	128	117	(16)
(17)		PrecMin	645	12	12	7	6	34	27	32	41	22	39	38	29	(17)
(18)		PrecStd	100	18	15	23	17	22	25	25	27	20	25	24	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.2	1.2	6.0	17.2	26.2	30.8	31.1	27.2	23.0	15.2	4.2	0.1	(19)
(20)			MCWB	-0.3	0.6	2.3	10.8	14.9	19.4	21.0	20.5	15.5	10.6	3.8	-0.2	(20)
(21)		2%	DB	-3.2	-1.8	3.2	12.2	22.2	28.2	29.0	25.0	20.0	11.1	2.1	-1.1	(21)
(22)			MCWB	-3.5	-2.2	0.6	7.4	12.5	18.5	19.6	18.2	14.5	8.1	1.9	-1.5	(22)
(23)		5%	DB	-5.9	-5.7	1.9	9.9	18.8	26.9	27.1	23.1	17.2	9.0	1.0	-2.9	(23)
(24)			MCWB	-6.1	-6.4	0.0	5.9	11.8	18.0	19.3	17.4	13.3	7.3	0.8	-3.2	(24)
(25)		10%	DB	-8.0	-7.8	0.7	7.8	15.1	24.3	25.2	21.1	15.0	7.1	-0.1	-5.0	(25)
(26)			MCWB	-8.3	-8.4	-0.6	4.5	9.6	17.2	18.6	16.7	11.9	5.9	-0.4	-5.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	0.7	2.6	10.8	15.7	21.4	22.8	21.6	17.0	10.9	3.9	0.0	(27)
(28)			MCDWB	-0.5	1.2	5.1	16.1	23.5	27.2	28.1	26.2	21.1	14.3	4.0	0.0	(28)
(29)		2%	WB	-3.8	-2.1	1.1	8.0	13.7	20.1	21.4	19.7	15.2	9.0	1.9	-1.8	(29)
(30)			MCDWB	-3.6	-1.8	2.4	11.7	20.7	26.1	26.7	23.1	18.5	10.5	2.1	-1.5	(30)
(31)		5%	WB	-6.3	-6.2	0.2	6.1	12.2	18.9	20.3	18.4	13.9	7.3	0.7	-3.3	(31)
(32)			MCDWB	-6.1	-5.6	1.4	9.1	17.8	24.8	25.1	21.7	16.8	8.4	0.9	-3.0	(32)
(33)		10%	WB	-8.4	-8.4	-0.9	4.5	10.2	17.9	19.4	17.2	12.4	5.9	-0.5	-5.5	(33)
(34)			MCDWB	-8.1	-7.8	0.5	7.1	15.1	23.2	23.8	20.4	14.8	6.9	-0.3	-5.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	8.0	9.2	9.0	9.4	9.8	8.8	7.4	5.5	6.6	7.4	(35)	
(36)			MCDBR	7.1	7.5	8.4	11.0	14.3	13.0	12.6	11.6	10.4	7.6	4.9	7.3	(36)
(37)			MCWBR	7.2	7.1	6.8	7.4	7.7	6.3	6.2	6.4	6.1	5.4	4.9	7.2	(37)
(38)		5% WB	MCDWB	7.2	7.6	8.0	10.3	13.4	11.4	11.1	10.6	9.6	7.2	4.9	7.3	(38)
(39)			MCWBR	7.3	7.3	6.8	7.3	8.1	6.1	5.9	6.5	6.1	5.4	5.0	7.3	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.259	0.302	0.354	0.374	0.387	0.402	0.437	0.410	0.361	0.329	0.268	0.245	(40)
(41)		taud		1.966	2.005	1.952	2.069	2.284	2.332	2.210	2.270	2.360	2.238	2.018	1.907	(41)
(42)		Ebn at Noon		431	640	729	806	834	826	779	769	747	622	414	269	(42)
(43)		Edh at Noon		45	82	123	133	117	113	125	109	83	66	42	30	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.40	1.29	2.82	4.43	4.85	5.44	5.12	3.55	2.23	0.96	0.39	0.20	(44)
(45)		RadStd		0.03	0.11	0.18	0.43	0.40	0.45	0.59	0.43	0.27	0.11	0.09	0.02	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page